The Dissonance Router

Evidence summary · Nebius x NVIDIA Global AI Hackathon · by Omkar Ramesh

The router sends each request to Nemotron Nano, Super or Ultra by how much a user's stated preferences contradict their observed behaviour. Below is what was measured on the real Nebius Token Factory models. Every figure comes from the saved reports in **docs/reports/** of the repository.

What was run

MEASURE	RESULT
Real Token Factory reports	**512 (433 clean, 49 failures, 30 fake claims)**
Local runs (Ollama, RTX 5090)	**31, kept as a separate group**
Scenarios routed as expected	**320 of 355 (90%)**
Blind Nano vs Ultra comparison	**87 cases: Ultra 56, Nano 6, tied 25**
Cost, router vs always-Ultra	**93% lower ($0.46 vs $6.32, planning prices)**
Unit tests	**315 passing**

Models

TIER	TOKEN FACTORY MODEL	USED WHEN
Nano	nvidia/NVIDIA-Nemotron-3-Nano-30B-A3B	Words and behaviour agree (score below 0.4); also reads each message for style requests
Super	nvidia/nemotron-3-super-120b-a12b	A preference is emerging (score 0.4 to 0.7)
Ultra	nvidia/Nemotron-3-Ultra-550b-a55b	Clear contradiction (score 0.7 or more), reasoning on

Where it fails (kept visible on purpose)

- The Nano reader misses some unusual phrasings of a style request. The held-out scenarios **ho_spell_it_out** and **ho_receipts** fail repeatedly.
- Local and Token Factory runs fail on different scenarios, so they are reported separately.
- In the demo session, the router saved only 8% against always-Ultra, because the demo escalates deliberately. The 93% figure is over the full test set.

How to read the cost figures

Costs use planning prices of $0.06 / $0.24 (Nano), $0.30 / $0.90 (Super) and $1.00 / $3.00 (Ultra) per million input / output tokens. These are planning figures and overstate the real bill; they have not yet been checked against Nebius billing. Predicted and measured per-answer costs differ by about 41% on average in the sampled session.

Repository: github.com/omkar-ramesh/The-dissonance-router · MIT licence · the README lists setup steps and the evaluation method.